

AI WasteWise

Problem Statement & Project Explanation

1M1B AI for Sustainability Virtual Internship

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1. Background

Waste segregation is an important part of responsible waste management. However, people may be uncertain about which category a particular item belongs to and how it should be disposed of. This uncertainty can make correct segregation difficult at the point where waste is generated.

2. Problem Statement

People often face difficulty identifying the correct category and disposal method for different types of waste. This uncertainty can lead to improper segregation and mixing of recyclable and non-recyclable materials. Students, households, offices, campus staff, and communities need a simple and accessible way to identify waste and receive appropriate disposal guidance.

AI WasteWise addresses this problem by using AI to analyze text or images of waste and provide a likely classification, disposal recommendation, confidence information, and a sustainability tip.

3. Why This Problem Matters

- Incorrect segregation can make waste handling more difficult.
- People may not know the appropriate disposal method for unfamiliar items.
- Simple, accessible guidance can help users make better decisions at the point of disposal.
- Improving awareness and segregation supports broader sustainability goals.

4. Affected Users

- **Students:** Need quick guidance for waste generated on campus.
- **Households:** Need an easy way to identify everyday waste.
- **Offices:** Can use simple guidance to support responsible disposal practices.
- **Campus staff:** Can benefit from clearer waste identification at source.
- **Communities:** Can use an accessible tool to improve sustainability awareness.

5. Root Causes of the Problem

- Lack of knowledge about waste categories.
- Uncertainty about disposal rules for specific or unfamiliar items.
- Difficulty identifying some waste items from appearance alone.

- Need for quick guidance instead of searching through lengthy information.

6. Limitations of a Manual Approach

A manual approach depends on the user's existing knowledge or on searching for disposal information separately. This can be time-consuming and may not provide an immediate answer for unfamiliar waste items. A digital assistant can bring identification and guidance together in one interaction.

7. Why AI Is Needed

- AI can analyze a user's text description of a waste item.
- AI can analyze an uploaded image and identify a likely waste type.
- AI can combine classification with practical disposal guidance.
- AI can provide a confidence indication and make uncertainty visible to the user.
- AI can deliver concise sustainability tips alongside the result.

8. Proposed Solution — AI WasteWise

AI WasteWise is a multimodal AI-powered waste segregation and recycling assistant. The user can either enter a waste item as text or upload an image. The application sends the input through its backend to Gemini AI for analysis. The system then presents the result in a simple format so that the user can understand what the item is likely to be and how it should be handled.

Input	AI Analysis	Output
Text description or image	Waste identification and classification	Waste type/category
User-provided item	Disposal decision support	Disposal guidance
AI result	Confidence/uncertainty handling	Confidence information
AI result	Sustainability guidance	Actionable sustainability tip

9. Basic Solution Workflow

- **Step 1 — Input:** User enters a waste item or uploads an image.
- **Step 2 — Frontend:** The web interface sends the input to the application backend.
- **Step 3 — Backend:** Node.js/Express handles the request and communicates with Gemini AI.
- **Step 4 — AI Analysis:** Gemini analyzes the input and generates the classification and guidance.
- **Step 5 — Result:** AI WasteWise displays the waste type, disposal guidance, confidence information, recommendation, and sustainability tip.

10. Responsible AI Considerations

- **Fairness:** Improve the system across different waste examples and contexts rather than assuming one example represents all cases.

- **Transparency:** Show the classification and confidence/verification signal so users understand that AI results may have uncertainty.
- **Privacy:** Avoid unnecessary collection of personal information and handle uploaded inputs responsibly.
- **Ethical Use:** Encourage safe and responsible disposal and avoid presenting uncertain classifications as guaranteed facts.

11. SDG Alignment

Primary SDG 12 — Responsible Consumption and Production: The project focuses on helping users make more responsible waste identification and disposal decisions.

Secondary SDG 11 — Sustainable Cities and Communities: Better waste practices can contribute to cleaner and more sustainable communities.

12. Expected Impact

- **Environmental:** Encourage better segregation at source and more responsible disposal decisions.
- **Social:** Make sustainability guidance easier to understand and access.
- **Technological:** Demonstrate the use of multimodal AI for a real-world sustainability problem.

Note: These are expected/planned impacts of the prototype, not measured field-study results.

13. Future Scope

- Improve image recognition across a wider range of waste types.
- Add localized recycling and disposal information.
- Introduce multilingual support.
- Develop a campus waste analytics dashboard.
- Extend the solution to mobile applications and smart waste-management systems.

14. One-Line Project Summary

AI WasteWise uses multimodal AI to help people identify waste and make better segregation and disposal decisions through simple, actionable guidance.